

Complex analysis, Lecture 1

What complex analysis is about

Complex analysis is the study of functions from $\mathbb{C}$ to $\mathbb{C}$ that can be differentiated.

These functions are known by the fancy name of “holomorphic functions”.

Definition

Let $\Omega \subseteq \mathbb{C}$ be open. A function $f : \Omega \rightarrow \mathbb{C}$ is **holomorphic** on Ω if at every $z_0 \in \Omega$ the limit

$$f'(z_0) := \lim_{h \rightarrow 0} \frac{f(z_0 + h) - f(z_0)}{h}, \quad h \in \mathbb{C} \setminus \{0\},$$

exists.

A function is called **complex analytic** if it is locally represented by a convergent power series. Every analytic function is holomorphic; the converse is a fundamental theorem we will prove later. Thus the two terms are ultimately interchangeable, although their definitions look different.

What is the main difference with real analysis?

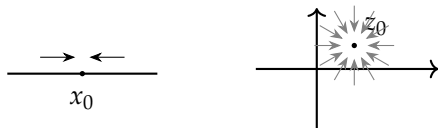
At first sight, the only difference with real analysis is that *taking a derivative is more constraining*.

On $\mathbb{R}$: two directions, two limits, required to match.

$$\lim_{h \rightarrow 0^-} \frac{f(x_0 + h) - f(x_0)}{h} = \lim_{h \rightarrow 0^+} \frac{f(x_0 + h) - f(x_0)}{h}.$$

On $\mathbb{C}$: left, right, top, bottom, and everything in between — all required to match.

$$\lim_{t \rightarrow 0} \frac{f(z_0 + t) - f(z_0)}{t} = \lim_{t \rightarrow 0} \frac{f(z_0 + it) - f(z_0)}{it} = \lim_{t \rightarrow 0} \frac{f(z_0 + e^{i\theta}t) - f(z_0)}{e^{i\theta}t}$$



This small difference is gigantic

As it turns out, this difference has a gigantic effect, and is *very* constraining. Two basic theorems to which we will come later:

Theorem (identity theorem)

Let f and g be holomorphic on open sets U and V , with $U \cap V$ connected. If $f = g$ on some nonempty open subset of $U \cap V$, then $f = g$ on all of $U \cap V$.

Corollary (isolated zeros)

Let f be holomorphic on a connected open set Ω . If $f(z_n) = 0$ for a sequence of distinct points $z_n \rightarrow z \in \Omega$, then $f \equiv 0$ on Ω .

Consequently (apply this to $f - c$) a holomorphic function on a connected open set that is constant on a nontrivial line segment contained in the set is constant everywhere. Nothing like this holds for C^∞ functions (think of a bump function): holomorphic is a much more restricted class.

Why should you care?

Despite this rigidity (or maybe because of it), complex analysis is a hugely useful field. It appears in the study of:

- ▶ combinatorics — *generating functions*,
- ▶ algebraic geometry — *Riemann surfaces*,
- ▶ fluid mechanics — *conformal maps, potential flow*,
- ▶ number theory, analytic and algebraic — *the zeta function, modular forms*,
- ▶ harmonic analysis — *the Fourier transform, Hardy spaces*,
- ▶ approximation theory — *rate of convergence of polynomial approximation*,
- ▶ probability — *Brownian motion, conformal invariance*,
- ▶ PDE and mathematical physics — *Laplace's equation, potential theory*,
- ▶ dynamical systems — *Julia and Mandelbrot sets*,
- ▶ engineering — *Laplace transform, control theory*,

and so on.

Back to the beginning: what does differentiability mean?

Let's start at the beginning and ponder the meaning of differentiability for holomorphic functions.

First let's reduce the study of complex differentiable functions to the case of functions from $\mathbb{R}^2$ to $\mathbb{R}^2$. Given $f : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ with

$$f(x, y) = (u(x, y), v(x, y)),$$

we think of it as a complex function by identifying it with

$$f(x + iy) = u(x, y) + i v(x, y).$$

The Cauchy–Riemann equations

Theorem

Let $f = (u, v)$ be differentiable at (x_0, y_0) as a map $\mathbb{R}^2 \rightarrow \mathbb{R}^2$, and $z_0 = x_0 + iy_0$. Then $f(x + iy) = u + iv$ is complex differentiable at z_0 if and only if

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}, \quad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x} \quad \text{at } (x_0, y_0).$$

The theorem is saying that for complex differentiability **only two directions matter**: top–bottom and left–right. If f is differentiable as a real map, the derivatives from these two directions determine all the others, and Cauchy–Riemann is exactly the condition for them to fit together into a complex derivative.

Proof of ($\Rightarrow$)

Suppose f is complex differentiable at z_0 . Then the limit defining $f'(z_0)$ can be taken along any direction; take two.

Left-right: $h = t \in \mathbb{R}, t \rightarrow 0$:

$$f'(z_0) = \lim_{t \rightarrow 0} \frac{f(z_0 + t) - f(z_0)}{t} = u_x + i v_x.$$

Top-bottom: $h = it, t \in \mathbb{R}, t \rightarrow 0$:

$$f'(z_0) = \lim_{t \rightarrow 0} \frac{f(z_0 + it) - f(z_0)}{it} = \frac{u_y + i v_y}{i} = v_y - i u_y.$$

The two must agree. Comparing real and imaginary parts,

$$u_x = v_y, \quad v_x = -u_y. \quad \square$$

Proof of ($\Leftarrow$)

Assume the Cauchy–Riemann equations hold at (x_0, y_0) .

Taylor expand, as real functions. Since u and v are differentiable at (x_0, y_0) , for $(h_1, h_2) \rightarrow (0, 0)$:

$$u(x_0 + h_1, y_0 + h_2) = u(x_0, y_0) + h_1 u_x + h_2 u_y + o(|h|),$$

$$v(x_0 + h_1, y_0 + h_2) = v(x_0, y_0) + h_1 v_x + h_2 v_y + o(|h|),$$

where $|h| = \sqrt{h_1^2 + h_2^2}$ and all partials are evaluated at (x_0, y_0) .

Reinterpret as a complex function. Put $h = h_1 + ih_2$, so that $z_0 + h \leftrightarrow (x_0 + h_1, y_0 + h_2)$. Then

$$\begin{aligned} f(z_0 + h) - f(z_0) &= [u(x_0 + h_1, y_0 + h_2) - u(x_0, y_0)] \\ &\quad + i [v(x_0 + h_1, y_0 + h_2) - v(x_0, y_0)] \\ &= h_1 (u_x + i v_x) + h_2 (u_y + i v_y) + o(|h|). \end{aligned}$$

Proof of ($\Leftarrow$), continued

We have

$$f(z_0 + h) - f(z_0) = h_1 (u_x + i v_x) + h_2 (u_y + i v_y) + o(|h|).$$

Use Cauchy–Riemann. Substituting $u_y = -v_x$ and $v_y = u_x$,

$$u_y + i v_y = -v_x + i u_x = i (u_x + i v_x).$$

So both coefficients are multiples of the same complex number $u_x + i v_x$:

$$f(z_0 + h) - f(z_0) = (h_1 + i h_2) (u_x + i v_x) + o(|h|) = h (u_x + i v_x) + o(|h|).$$

Conclude. Divide by h :

$$\frac{f(z_0 + h) - f(z_0)}{h} = (u_x + i v_x) + \frac{o(|h|)}{h} \longrightarrow u_x + i v_x$$

as $h \rightarrow 0$ from any direction, since $|o(|h|)/h| = o(|h|)/|h| \rightarrow 0$. So f is complex differentiable at z_0 , with $f'(z_0) = u_x + i v_x$. $\square$

What is the meaning of the Cauchy–Riemann equations?

The simplest interpretation is to consider a 2-dimensional fluid with velocity field

$$(x, y) \longmapsto (u(x, y), v(x, y)).$$

Recall the divergence and the curl of a planar vector field:

$$\operatorname{div}(u, v) := \frac{\partial u}{\partial x} + \frac{\partial v}{\partial y}, \quad \operatorname{curl}(u, v) := \frac{\partial v}{\partial x} - \frac{\partial u}{\partial y}.$$

In two dimensions the curl is a scalar: it is the z-component of the three-dimensional curl $\nabla \times (u, v, 0)$, the only component that can be nonzero for a planar field.

Cauchy–Riemann, in the language of flows

The pair $(u, -v)$ satisfies the Cauchy–Riemann equations if and only if

$$\operatorname{div}(u, v) = 0 \quad \text{and} \quad \operatorname{curl}(u, v) = 0.$$

The minus sign is forced: Cauchy–Riemann for (u, v) itself would say $u_x = v_y$ and $u_y = -v_x$, which is neither $\operatorname{div} = 0$ nor $\operatorname{curl} = 0$; flipping the sign of v turns them into exactly those two.

In terms of fluids: $\operatorname{div} = 0$ means the flow is **incompressible**, and $\operatorname{curl} = 0$ means it is **irrotational**. Let's explain.

Incompressible and irrotational

Write $V = (u, v)$, and let D be any small region with boundary ∂D .

Incompressible: $\operatorname{div} V = 0$. This is *conservation of mass* for a fluid of constant density. By the divergence theorem,

$$\oint_{\partial D} V \cdot n \, ds = \iint_D \operatorname{div} V \, dA = 0,$$

so as much fluid flows into D as flows out. No sources, no sinks.

Irrotational: $\operatorname{curl} V = 0$. By Green's theorem,

$$\oint_{\partial D} V \cdot t \, ds = \iint_D \operatorname{curl} V \, dA = 0,$$

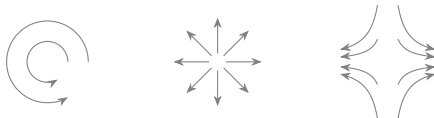
so the *circulation* around ∂D vanishes: the fluid does not swirl around any point.

Upshot: holomorphic functions $f = u - iv$ are the same thing as incompressible, irrotational plane flows.

Which flows come from holomorphic functions?

This allows us to immediately see which vector fields (u, v) do **not** correspond to holomorphic functions.

- ▶ **Rotation.** $(u, v) = (-y, x)$: the fluid circulates in closed loops around the origin. $\text{curl}(u, v) = 1 - (-1) = 2 \neq 0$: vorticity. Indeed $u - iv = -i\bar{z}$ is not holomorphic.
- ▶ **Source.** $(u, v) = (x, y)$: fluid pours out of the origin. $\text{div}(u, v) = 2 \neq 0$. Indeed $u - iv = \bar{z}$ is not holomorphic.
- ▶ **Stagnation point.** $(u, v) = (x, -y)$: $\text{div} = 1 - 1 = 0$, $\text{curl} = 0 - 0 = 0$. This one is fine: $u - iv = x + iy = z$.



A standard fact from vector calculus

Standard fact (Poincaré lemma on a disk)

Let D be an open disk and $W = (W_1, W_2)$ a C^1 vector field on D with $\operatorname{curl} W = 0$.
Then W is a gradient:

$$W = \nabla \varphi \quad \text{for some } \varphi : D \rightarrow \mathbb{R},$$

unique up to an additive constant.

The two potentials

Let $f = u - iv$ be holomorphic on a disk D , $u, v \in C^1$, the associated flow being $V = (u, v)$: curl-free and divergence-free.

Velocity potential. $\text{curl}(u, v) = 0$, so there is φ with

$$\nabla \varphi = (u, v), \quad \text{i.e.} \quad \varphi_x = u, \quad \varphi_y = v.$$

Stream function. Rotate by 90° : $\text{curl}(-v, u) = u_x + v_y = \text{div}(u, v) = 0$, so there is ψ with

$$\nabla \psi = (-v, u), \quad \text{i.e.} \quad \psi_x = -v, \quad \psi_y = u.$$

Put them together. Comparing, $\varphi_x = \psi_y$ and $\varphi_y = -\psi_x$: these are the Cauchy–Riemann equations for $F := \varphi + i\psi$. So F is holomorphic on D , and

$$F' = \varphi_x + i\psi_x = u - iv = f.$$

Derivatives of holomorphic functions

We have shown that every holomorphic function on a disk has a holomorphic primitive. We will also use that the derivative of a holomorphic function is holomorphic. This is a deeper theorem in general, but easy under slightly stronger regularity:

Proposition

Let $f = \varphi + i\psi$ be holomorphic with $\varphi, \psi \in C^2$. Then $f' = \varphi_x + i\psi_x$ is holomorphic, with C^1 parts.

Derivatives of holomorphic functions: proof

Proof. Differentiate the Cauchy–Riemann equations of f , using equality of mixed partials:

$$(\varphi_x)_x = (\psi_y)_x = (\psi_x)_y, \quad (\varphi_x)_y = (\varphi_y)_x = (-\psi_x)_x = -(\psi_x)_x.$$

These are the Cauchy–Riemann equations for $\varphi_x + i\psi_x$. $\square$

Standing assumption from now on: holomorphic functions have C^2 (in practice, smooth) real and imaginary parts. That this is automatic is a theorem for later.

Holomorphic functions as flows

Take any holomorphic $f = \varphi + i\psi$, and consider the flow with velocity field

$$V = \nabla \varphi = (\varphi_x, \varphi_y).$$

By Cauchy–Riemann, $\varphi_x - i\varphi_y = \varphi_x + i\psi_x = f'$, so this is the flow associated to f' in the sense of the earlier slides.

Vocabulary. f is the **complex potential** of the flow and f' its **complex velocity**. The velocity vector is $V = \overline{f'}$ and the speed is $|f'|$.

Two things to explain:

- ▶ the curves $\{\operatorname{Im} f = c\}$ are the **streamlines**: the paths the particles follow;
- ▶ the **flux** across a curve from A to B is $\operatorname{Im} f(B) - \operatorname{Im} f(A)$.

Streamlines: the level sets of $\operatorname{Im} f$

Setup. $f = \varphi + i\psi$ holomorphic, velocity $V = \nabla \varphi = (\varphi_x, \varphi_y)$. By Cauchy–Riemann, $\nabla \psi = (\psi_x, \psi_y) = (-\varphi_y, \varphi_x)$ is V rotated by 90° .

A **streamline** is the path of a fluid particle: a curve $\gamma(t)$ whose velocity is that of the fluid,

$$\gamma'(t) = V(\gamma(t)) = \nabla \varphi(\gamma(t)).$$

Claim: ψ does not change along a streamline. By the chain rule,

$$\frac{d}{dt} \psi(\gamma(t)) = \nabla \psi \cdot \gamma'(t) = \nabla \psi \cdot \nabla \varphi = (-\varphi_y, \varphi_x) \cdot (\varphi_x, \varphi_y) = 0.$$

So ψ is a *conserved quantity* of the motion: a particle starting on $\{\psi = c\}$ stays on $\{\psi = c\}$.

Level sets of $\operatorname{Im} f$ are the streamlines of the flow of f' .

How much fluid crosses a curve?

Pick a holomorphic $f = \varphi + i\psi$, with its flow $V = \nabla \varphi$ (the flow described by f'), and a curve C from A to B .

Question. How much fluid passes through C per unit time? This is the **flux** of V across C : with (dx, dy) the tangent of C and $n ds := (dy, -dx)$ the normal,

$$\text{flux} = \int_C V \cdot n ds = \int_C \varphi_x dy - \varphi_y dx.$$

Proposition

The flux across C is $\text{Im} f(B) - \text{Im} f(A)$: it depends only on the endpoints.

Proof. By Cauchy–Riemann, $\varphi_x = \psi_y$ and $-\varphi_y = \psi_x$, so the integrand is $\psi_y dy + \psi_x dx = d\psi$, and $\int_C d\psi = \psi(B) - \psi(A)$. $\square$

Consequence. Between two streamlines $\{\psi = c_1\}$ and $\{\psi = c_2\}$ the flow rate is $c_2 - c_1$, wherever you cut across. So where streamlines come closer together, the fluid moves faster.

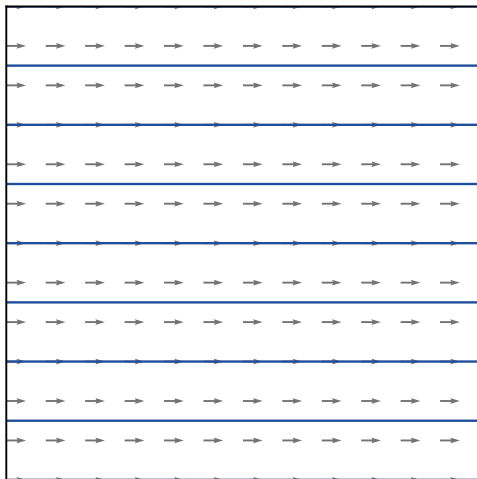
Example 1: $f'(z) = 1$

The flow. $f' = 1 = \varphi_x - i\varphi_y$: velocity field $V = (1, 0)$. Every particle moves to the right at unit speed.

The primitive. $f(z) = z = x + iy$, so $\varphi = x$, $\psi = y$. Check: $\nabla\varphi = (1, 0) = V$.

The streamlines. $\{\operatorname{Im}f = c\} = \{y = c\}$: horizontal lines, as they should be.

The flux. Across the vertical segment from $A = (0, 0)$ to $B = (0, h)$: speed 1 along a segment of length h , so the flux is h . And $\operatorname{Im}f(B) - \operatorname{Im}f(A) = h - 0 = h$.



Example 2: $f'(z) = z$

The flow. $f' = x + iy = \varphi_x - i\varphi_y$, so the velocity at (x, y) is

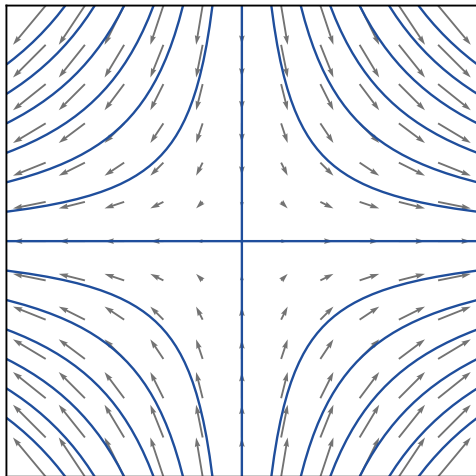
$$V = (x, -y).$$

Away from the y -axis horizontally, toward the x -axis vertically. At the origin the velocity is 0: a **stagnation point**.

The primitive.

$$f(z) = \frac{1}{2}z^2 = \frac{1}{2}(x^2 - y^2) + ixy. \text{ Check:}$$
$$\nabla\varphi = (x, -y) = V.$$

The streamlines. $\{\operatorname{Im} f = c\} = \{xy = c\}$: hyperbolas with the axes as asymptotes. A particle comes in along the y -axis direction, turns, and leaves along the x -axis direction.



Example 3: $f'(z) = 1/z$

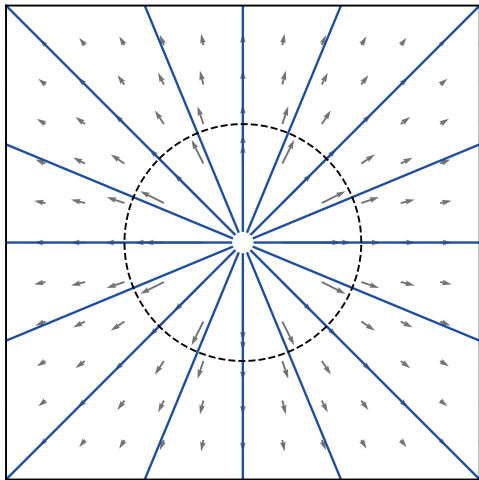
The flow. $f' = \frac{1}{z} = \frac{x - iy}{x^2 + y^2}$, so

$$V = \frac{(x, y)}{r^2}, \quad r^2 = x^2 + y^2.$$

Radially outward, speed $1/r$: a **source** at the origin.

A primitive, on a disk not containing 0: choose a continuous branch of the angle and set $f(z) = \log z = \log r + i\theta$. Check: $\nabla \log r = (x, y)/r^2 = V$.

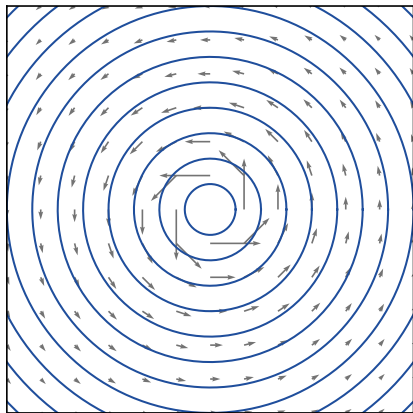
The streamlines. $\{\theta = c\}$: rays from the origin.



Example 3: $1/z$ has a singularity at 0

The flux out of a circle. Across $|z| = \rho$ the fluid moves outward at speed $1/\rho$ along a curve of length $2\pi\rho$: the flux is $2\pi \neq 0$. Fluid is being *created* at the origin.

Why the singularity is not removable. If $1/z$ had a holomorphic extension to a disk containing the circle, it would have a primitive f on that disk (the two potentials), and the flux across the circle would be $\text{Im}f(B) - \text{Im}f(A)$ with $A = B$: zero. It is 2π . So there is no such disk: the origin is a genuine singularity.



The vortex $f' = -i/z$: circles as streamlines, circulation 2π around the origin. Same story, with flux replaced by circulation.

Example 3: no global primitive on $\mathbb{C} \setminus \{0\}$

$\operatorname{Im} \log z = \theta$ increases by 2π around the circle: $\log z$ is not single-valued on $\mathbb{C} \setminus \{0\}$, and any other primitive would differ from it by a constant. So $1/z$ has **no primitive on $\mathbb{C} \setminus \{0\}$** , which is what the flux computed. This is why the standard fact was stated on a disk.

Example 4: $f(z) = z + 1/z$ — flow past a cylinder

This time start from the primitive.

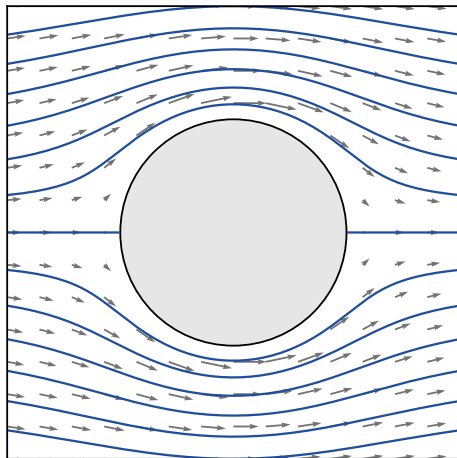
$f(z) = z + 1/z$ is holomorphic on $\mathbb{C} \setminus \{0\}$,
and

$$\operatorname{Im} f = y - \frac{y}{x^2 + y^2} = y \left(1 - \frac{1}{r^2} \right).$$

The streamlines are the curves

$$y \left(1 - \frac{1}{r^2} \right) = c.$$

- ▶ $c = 0$ is the unit circle $r = 1$ (with the x -axis). So **the unit circle is a streamline**: no fluid crosses it. The flow does not notice whether the disk is full of fluid or a solid cylinder.
- ▶ For large r , $\operatorname{Im} f \approx y$: far away, the uniform flow of Example 1.



Example 4, continued: where is the fluid fast?

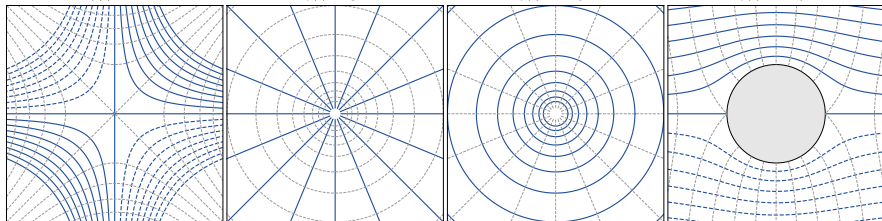
The velocity is $\overline{f'(z)}$, with $f'(z) = 1 - \frac{1}{z^2}$; the speed is $|f'(z)|$. On the circle $z = e^{i\theta}$,
 $|f'| = |1 - e^{-2i\theta}| = 2 |\sin \theta|$.

- ▶ It vanishes at $z = \pm 1$: **stagnation points**, at the front and back of the cylinder. The fluid comes to rest there and splits to go around.
- ▶ At the top, $z = i$: $f'(i) = 1 - 1/i^2 = 2$. Likewise at the bottom. The fluid moves at **twice** the far-field speed there.

Look at the picture: above and below the cylinder the streamlines are squeezed together. That is the flux proposition at work — the same amount of fluid has to pass through a narrower channel, so it speeds up.

Four flows

Streamlines $\{\operatorname{Im} f = c\}$ in blue; level sets $\{\operatorname{Re} f = c\}$ dashed in grey.



- ▶ z^2 : flow into a corner. Stagnation point at 0: streamlines spread apart, speed 0.
- ▶ A branch of $\log z$: source. A branch of $-i \log z$: vortex. Neither extends holomorphically across 0: the flux out of (resp. circulation around) a circle about 0 is $2\pi \neq 0$, impossible for an incompressible, irrotational flow on a disk containing 0.
- ▶ $z + 1/z$: uniform flow past the unit disk; the circle $|z| = 1$ is itself a streamline ($\psi = 0$).